Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Hazard Analysis

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1.0 Purpose and Scope

This document is the preliminary hazard analysis of the System: its hazards, their causes and severity, and the controls proposed against them.

The hazard analysis is delivered once per safety phase, each closed by a safety review [2, Secs. 3.1 and 4]:

Phase 0 Concept complete: the top hazards and their causes.
Phase 1 Requirements nearly complete: all known hazards, with their causes, controls and risk.
Phase 2 Design mostly complete: the approved controls, and how each is verified.
Phase 3 Testing complete: proof that the verifications passed.

This document is the first delivery: phases 0 and 1, which small projects combine [2, Sec. 4]. The controls are proposals that the requirements turn into rules; each already names its verification method [7, App. F.3, p. 348].

It covers the software items of the Software Classification [6, Sec. 3.0] in operation: all controller modes, during and between contacts. Bench and build tools are covered where they can reach operation. Its hazards decide which software is safety-critical; the Software Classification records that designation [6, Sec. 4.0].

How to read it. Section 4.0 lists every hazard, how severe it is, and whether it is tolerant enough. Section 5.0 explains in plain words why some are not, and what has to change. Section 6.0 gives one report per catastrophic or critical hazard. The appendices hold the analyses behind them: the fault tree, the failure tables, the glossary and the open items.

2.0 Documents

[1] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022. Causes cite App. A, Table 2, pp. 62–68.
[2] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 8.58 “Software Safety and Hazard Analysis”.
[3] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 8.05 “SW Failure Modes and Effects Analysis”.
[4] NASA, *Fault Tree Handbook with Aerospace Applications*, Version 1.1, August 2002.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.

[6] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Classification*, Issue 1, Revision 1, Draft.
[7] NASA, *NASA Software Safety Guidebook*, NASA-GB-8719.13, 2004.
[8] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016.
[9] NASA, *Fault Management Handbook*, NASA-HDBK-1002, Draft 2, 2 April 2012. A draft, not a released standard.

3.0 Ground Rules

These rules set the boundary, the failures included, the fault tolerance required and how failures are modelled and named, before the analysis starts [3, Sec. 3.1; 4, pp. 23–24].

3.1 Boundary

- **Harm:** the loss of the controller's operability, observability or correctability.
- **Boundary:** the radio equipment and the spacecraft power supply lie outside the System [5, Sec. 1.2.2]. Their failures enter as causes and are not analysed further.
- **Physical tampering:** excluded; nobody can reach the spacecraft (Assumption A-1).

3.2 Failures Included

The failures below are taken from the software causes of hazards [1, App. A, Table 2, pp. 62–68].

Domain	Failures
Software	Hang, crash, buffer overflow or invalid state in any software item
Data	Message corrupted, lost, repeated or late on any link; full buffer
Memory	Write cut short by power loss; single-event upset (a bit flipped by radiation, A-5); corrupted value that survives a restart
Timing	Watchdog timeout; restart loop; onboard clock wrong, drifting or reset
Power	Unannounced loss at any moment (A-2)
Hardware link	Physical failure of a data-link connection or the reprogramming link
Operator	Invalid, out-of-sequence or mistimed telecommand (A-4)
Security	Means of proving origin disclosed

3.3 Analysis Rules

This section defines the evaluation rules, fault tolerance criteria, and baseline assumptions governing the formal hazard and fault tree analyses [3, Sec. 3.1; 7, Sec. 3.2.1].

- **Counting failures:** to check a hazard, we count the failures that would have to happen for it to occur. If one cause disables several controls at once, it counts as one failure, not one per control. Counting each control separately would make the System look more tolerant than it is.
- **Severity:** each hazard is rated by how bad its outcome is if it happens, using the four categories in the legend below. The rating ignores the controls: they make a hazard less likely, but its outcome is just as bad when it does happen. A hazard therefore keeps its severity however many controls it has.
- **Fault tolerance:** how many failures must happen together before a hazard can occur. A catastrophic hazard must need at least three of the failures in Section 3.2 together, a critical hazard at least two [3, Sec. 3.1]. For each hazard, we find the smallest set of failures that causes it and compare it with this number. A hazard that needs fewer breaks the rule, and its controls must be strengthened.

- **Safety effort:** full [7, Sec. 3.2.1, p. 44], because between contacts no operator can respond to a hazard (A-3). It decides the analyses here: a fault tree is mandatory and a failure modes analysis highly recommended [7, Table 6-3, p. 107].
- **Assumptions:** the analysis relies on the ConOps assumptions A-1 to A-5, and adds two of its own, each taken as the worst case:
  1. the last-resort firmware the restore puts back has the same start-up code, defects and kept data as the other images, so a cause that stops every image also stops the restored one;
  2. one means proves that both commands and firmware come from the operator, and both the onboard computer and the controller use it, so disclosing it opens both.

Legend: severity (categories from [7, Table 2-2, p. 27])

CATASTROPHIC	The controller is lost for the rest of the mission: it cannot run, or can no longer be commanded, observed or corrected.
CRITICAL	The controller acts against the operator's intent, or cannot be commanded, observed or corrected until the operator acts at a later contact.
MODERATE	Part of a function is lost, such as some health reports or one update attempt; the controller still runs and obeys the operator.
NEGLIGIBLE	Nothing is lost; the operator has extra work or gets data a contact later.

3.4 Naming

- **Hazards:** H-1 to H-17, in order of severity.
- **Fault tree:** gates G01 to G06; primary events B01 to B10. No identifier is ever reused.
- **ConOps numbers:** objectives (O), assumptions (A), constraints (C), risks (R), capabilities and scenarios refer to the ConOps [5, Secs. 1.1.2, 3.1, 3.5, 6.0 and 8.0].
- **Software role** [2, Sec. 3.2.2]: CAUSE, a software fault can lead to the hazard; CONTROL, software prevents it; MITIGATION, software limits its effect; MONITOR, software detects and reports it.
- **Verification method** [8, Sec. 5.3.1.2.1, p. 92]: TEST, using the product to obtain detailed data; DEMONSTRATION, using it to show the control works, without detailed data; INSPECTION, examining the product or its records; ANALYSIS, modelling or calculation.

4.0 Preliminary Hazard List

The hazards in order of severity. For each catastrophic and critical hazard, the table also gives how many failures must happen together to cause it, with the proposed controls in place, and whether that meets the fault tolerance rule (Section 3.3).

ID	Hazard	Severity	Failures to cause it	Meets rule
H-1	The controller is left with no firmware it can run, and cannot be restored	CATASTROPHIC	1	NO
H-2	Firmware from someone other than the operator runs on the controller	CATASTROPHIC	1	NO
H-3	New firmware can no longer reach the controller	CATASTROPHIC	2	NO
H-4	The controller can no longer be commanded or observed	CATASTROPHIC	1	NO
H-5	Someone other than the operator commands the System	CATASTROPHIC	1	NO

ID	Hazard	Severity	Failures to cause it	Meets rule
H-6	A command is acted on in a form or at a time the operator did not intend	CRITICAL	1	NO
H-7	Wrong firmware runs on the controller: damaged, out of date, or with a hidden defect	CRITICAL	2	YES
H-8	The controller stops working between contacts and is not brought back	CRITICAL	1	NO
H-9	The ground is shown a false state of the System	CRITICAL	1	NO
H-10	A fault between contacts never reaches the ground	CRITICAL	2	YES
H-11	A valid command is not acted on, and the ground is not told	CRITICAL	2	YES
H-12	A test function acts outside a test	CRITICAL	2	YES
H-13	A fault response is taken when there is no fault	MODERATE	N/A	N/A
H-14	An update cannot be completed	MODERATE	N/A	N/A
H-15	Health reports are lost	MODERATE	N/A	N/A
H-16	Reports reach the ground later than planned	NEGLIGIBLE	N/A	N/A
H-17	A command is refused with a wrong or unclear reason	NEGLIGIBLE	N/A	N/A

Legend. Failures to cause it: the smallest number of failures that, happening together, make the hazard occur. Meets rule: YES when this is at least three for a catastrophic hazard or two for a critical one; NO otherwise. N/A: the rule does not cover moderate and negligible hazards. Section 5.1 names the failures.

5.0 Findings

Eight of the twelve catastrophic and critical hazards do not meet the fault tolerance rule yet: seven can happen from a single failure, and H-3 from two.

5.1 What Causes Each Hazard

The smallest set of failures that, happening together, causes each catastrophic and critical hazard, with the proposed controls in place:

H-1 One failure: a fault that every firmware copy shares, including the one the restore puts back: the start-up code, the data kept over restarts, or a common defect (B08 to B10).

H-2 One failure: the means of proving origin is disclosed; both checks rely on it.

H-3 Two failures: a fault in the onboard computer's update path or its stored data, and its correction by the system outside this project failing (A-6).

H-4 One failure: a software defect that cuts both data-link connections, or the onboard computer failing for good.

H-5 One failure: the means of proving origin is disclosed.

H-6 One failure: a wrong onboard clock runs a scheduled command at the wrong time, or a bit error changes a held command after its check.

H-7 Two failures: a hidden defect in new firmware, and a trial that does not exercise it.

H-8 One failure: the same defect in every firmware copy; every restart, and the restore, brings it back.
H-9 One failure: a wrong onboard clock gives every report a wrong time.
H-10 Two failures: a fault, and fault detection that misses it.
H-11 Two failures: a command that is lost, and a ground that does not notice its answer never came.
H-12 Two failures: fault-injection functions left in the operating software, and a wrong command that starts them.

5.2 Shared Causes That Break Fault Tolerance

Each hazard above that breaks the rule does so through a cause that several controls share: one event defeats them all at once. The analysis looked for such shared causes, including the same defect in software that runs in more than one place [1, App. A.1.1, p. 62]. Each is listed with the hazards it reaches and what the requirements must change.

- **Shared image code and data.** Every firmware copy, including the last-resort firmware the restore puts back, shares the start-up code, the data kept over restarts and any defect they have in common (assumption 1). One fault there stops them all. Breaks H-1 and H-8; also reaches H-6 and H-13. *To change:* the last-resort firmware and the restore must not depend on what the other copies share.
- **One proof of origin.** One means proves that both commands and firmware come from the operator (assumption 2). If it is disclosed, another party can command the controller and load firmware. Breaks H-2 and H-5. *To change:* a disclosure must be noticed, so the means can be replaced in flight (Scenario-370).
- **Onboard computer software.** It is corrected by a system outside this project (A-6), so a defect in its update path stays until that system corrects it. Breaks H-3, which then needs two failures; also reaches H-1 and H-4. *Outside this project:* the outside system must be able to correct the onboard computer's update path and stored data.
- **One onboard computer.** It is the only path for commands, reports and restores; if it fails for good, all three stop (R-4). Breaks H-4; also reaches H-1, H-8 and H-10. *Outside this project:* how many onboard computers the spacecraft carries is a spacecraft decision; the risk is accepted (R-4).
- **Same data-link software.** The two data-link connections share software at both ends, so one defect cuts both (R-1) [1, p. 66]. Breaks H-4. *To change:* one defect must not be able to cut both connections.
- **Wrong onboard clock.** Expiry, arm time-outs, scheduled commands, time-tags and the trial time limit all depend on it [1, p. 65]. A wrong clock alone runs a scheduled command at the wrong time and gives reports wrong times. Breaks H-6 and H-9; also reaches H-4, H-11 and H-13. *To change:* no check may depend on the onboard clock alone.
- **Radiation.** A bit error can change a held command after it was checked and before it runs (A-5) [1, p. 63]. Breaks H-6; bit errors also reach H-1, H-7, H-8, H-9 and H-10. *To change:* a held command is checked again just before it runs.

5.3 Other Shared Causes

These defeat several controls at once, but no hazard falls to them alone.

- **One power supply.** It feeds both the onboard computer and the controller [5, Sec. 3.3]; one loss stops the controller and the computer that would restore it. Reaches H-1, H-8, H-10, H-11 and H-15.
- **One set of build tools.** The same tools build every firmware copy; a tool defect enters all of them. Reaches H-1, H-7 and H-12.

6.0 Hazard Reports

Reports cover the catastrophic and critical hazards, the two categories of the hazard report form [7, Fig. F-1, p. 351]; the others stay in the preliminary hazard list. Each report gives when the hazard can occur, its worst effect, its causes, the proposed controls and how they fail [2, Sec. 4].

In each report, the first table lists the causes. Source is the ConOps item or software cause it comes from, or NEW when this analysis found it. Controls gives the numbers of the controls that cover it; NONE marks a cause with no control yet, against the rule of at least one control for each cause [7, App. F.3, p. 348]. The second table lists the controls, each with its source and its verification method (Section 3.4). The shared causes behind the reports are in Section 5.0.

H-1 The controller is left with no firmware it can run, and cannot be restored. CATASTROPHIC. Software: CAUSE, CONTROL.
When: An update, a trial, any restart, or a needed restore.
Effect: The controller is lost for the rest of the mission (A-1).

	Cause	Source	Controls
a	Power lost while firmware is written	Scenario-600; [1, p. 63]	1
b	A bit error in stored firmware	Scenario-610; [1, p. 63]	2
c	An update writes over the previous or last-resort firmware	[1, p. 65]	1
d	New and previous firmware both fail	Scenario-460	4, 5
e	The last-resort firmware has changed unnoticed	[1, p. 63]	6
f	A restart loop never reaches the fall-back	[1, p. 64]	3, 7, 8
g	The start-up code fails or is damaged	[1, p. 63]	**NONE**
h	The reprogramming link fails, or the onboard computer misses the need to restore	Scenario-520	6
i	The onboard computer's known-good copy is damaged	[1, p. 63]	6

	Control	Source	Verified by
1	An update never writes over the previous or last-resort firmware	Scenario-600	TEST
2	Every image is checked before it runs	Scenario-610	TEST
3	After a set number of failed starts, the controller returns to its previous firmware	Scenario-440	TEST
4	A last-resort firmware that updates never replace	Scenario-460	TEST
5	A restore of the last-resort firmware through the reprogramming link, independent of the controller's software	Scenario-480	TEST
6	Stored firmware and the reprogramming link are checked between uses	NEW	TEST
7	A power loss during the trial counts as a failed start	Scenario-605	TEST
8	Failed firmware is not started again unless the operator orders it	Scenarios 430–460	TEST

Controls fail if: The image check passes damaged firmware; the failed-start count is lost at a restart; the last-resort firmware, or the image the restore puts back, shares the defect.

H-2 Firmware from someone other than the operator runs on the controller. CATASTROPHIC. Software: CONTROL.
When: An update or a restore.
Effect: Hostile firmware runs the controller and can refuse updates and report a false state. The restore starts only when no firmware can run (Scenario-480), so this firmware is never removed.

	Cause	Source	Controls
a	Firmware not from the operator arrives	Scenario-410; [1, p. 67]	1, 2, 3
b	The means of proving origin is disclosed	NEW	3, 4
c	A restore starts a stored copy someone else replaced	NEW	2, 3

	Control	Source	Verified by
1	The onboard computer refuses firmware not released by the operator	Scenario-410	TEST
2	The controller checks the origin again before it runs the firmware	NEW	TEST
3	The operator can order a restore while firmware runs	NEW	DEMONSTRATION
4	A disclosed means of proving origin is replaced in flight	Scenario-370	TEST

Controls fail if: The check accepts what it should refuse; the means of proof is disclosed without the operator knowing, so it is never replaced.

H-3 New firmware can no longer reach the controller. CATASTROPHIC. Software: CAUSE, CONTROL.
When: After an update is confirmed, or after a fault in the onboard computer.
Effect: The controller can no longer be corrected for the rest of the mission.

	Cause	Source	Controls
a	Confirmed firmware runs, but cannot receive firmware	NEW	1, 2, 3, 4
b	A defect in the onboard computer's update path	NEW	5
c	A wrong stored minimum version or controller identity refuses all firmware	[1, p. 67]	5
d	Firmware storage on the onboard computer stays full	[1, p. 66]	5

	Control	Source	Verified by
1	New firmware runs a trial before it is confirmed	Scenarios 430–450	TEST
2	It is judged good only after it has received firmware itself	NEW	TEST
3	The last-resort firmware can always receive firmware	NEW	TEST
4	The operator can order a return to the last-resort firmware	NEW	DEMONSTRATION
5	The system outside this project corrects the onboard computer's software and stored data	A-6	ANALYSIS

Controls fail if: The trial skips the update path; the outside system cannot correct the onboard computer.

H-4 The controller can no longer be commanded or observed. CATASTROPHIC. Software: CAUSE, CONTROL.
When: At any time.
Effect: The controller runs on its current firmware, but nobody can command or observe it for the rest of the mission.

	Cause	Source	Controls
a	Both data-link connections fail	Scenario-510	1, 2
b	One software defect cuts both connections	R-1; [1, p. 66]	**NONE**
c	A controller that never stops sending blocks the data link	[1, p. 63]	3
d	A wrong onboard clock makes every command expire, including the clock correction	[1, p. 65]	4
e	The ground loses the means of proving origin	NEW	**NONE**
f	A false fault response marks both connections failed	H-13	2, 5
g	A replacement of the means of proving origin goes wrong, so no command is accepted	Scenario-370	**NONE**
h	The onboard computer fails for good	R-4	**NONE**

	Control	Source	Verified by
1	Two separate data-link connections; one is enough	O-3; Scenario-500	TEST
2	A lost connection is reported	Scenario-510	TEST
3	The reprogramming link stays usable	Scenario-510	TEST
4	The clock correction does not depend on the onboard clock	NEW	TEST
5	A connection that works again is used again	Scenario-505	TEST

Controls fail if: One cause cuts both connections.

H-5 Someone other than the operator commands the System. CATASTROPHIC. Software: CONTROL.
When: During a contact, or at any time for a source on the spacecraft.
Effect: Another party can do anything the operator can, and hide it, for the rest of the mission.

	Cause	Source	Controls
a	A command not from the operator arrives	Scenario-300; [1, p. 67]	1
b	The means of proving origin is disclosed	NEW	3
c	Someone else uses the operator interface, and no second person checks a command	A-4	2
d	A command arrives from an unexpected system on the spacecraft	[1, p. 62]	1

	Control	Source	Verified by
1	Every command is proven to come from the operator; any other is refused and logged	O-2; Scenario-300	TEST
2	Only the operator can use the operator interface	NEW	DEMONSTRATION
3	A disclosed means of proving origin is replaced in flight	Scenario-370	TEST

Controls fail if: The means of proof is disclosed without the operator knowing, so it is never replaced.

H-6 A command is acted on in a form or at a time the operator did not intend. CRITICAL. Software: CAUSE, CONTROL.
When: While a command is checked, held or run.
Effect: The controller does what the operator did not intend; the ground learns of it at the next contact.

	Cause	Source	Controls
a	A command is changed in transfer	[1, p. 62]	1
b	An old command is sent again	Scenario-300	1
c	A command is run twice after a lost answer	Scenario-350	3
d	A command not allowed in the current mode arrives	Scenario-310; [1, p. 63]	1
e	A command with impossible values arrives	Scenario-320; [1, p. 62]	1
f	A dangerous command arrives without a valid arm	Scenarios 330, 335; [1, p. 63]	2, 6
g	A command arrives past its expiry	Scenario-340; [1, p. 63]	1
h	A scheduled command runs at the wrong time, or again after a restart	[1, p. 64]	**NONE**
i	A bit error changes a held command or the arm state	[1, p. 63]	**NONE**
j	The ground software builds a wrong command, or the operator enters one	A-4; [1, p. 63]	4, 5
k	A command of a sequence runs after an earlier one was refused or not answered	NEW	7

	Control	Source	Verified by
1	The onboard computer checks that every command comes from the operator, is new and has not expired; the controller checks that it is allowed in the current mode and has possible values	Scenario-120	TEST
2	The controller runs a dangerous command only after an arm, which expires	Scenarios 330, 335	TEST
3	A repeated command is answered “already done”	Scenario-350	TEST
4	The ground software checks every command before it is sent	Scenario-120	TEST
5	Every command and answer is logged	Scenario-120	INSPECTION
6	The onboard computer passes a dangerous command on only after an arm, which expires	Scenarios 330, 335	TEST
7	The ground software sends each command of a sequence only after the previous one is answered done, and stops at the first refusal	NEW	TEST

Controls fail if: The checks use a wrong clock; a held command or the arm state changes after its check; new firmware checks commands wrongly (H-7).

H-7 Wrong firmware runs on the controller: damaged, out of date, or with a hidden defect. CRITICAL. Software: CAUSE, CONTROL.
When: During and after an update, a fall-back or a restore.
Effect: The controller misbehaves until a further update. If the firmware passes its trial, it becomes the known-good firmware the controller falls back to. If it stops updates, H-3 follows.

	Cause	Source	Controls
a	A hidden defect the trial does not exercise	NEW	1, 5
b	Firmware changed after its check, in transfer or in storage	[1, p. 63]	2, 3
c	Firmware older than the minimum version is sent again	Scenario-420	2
d	The wrong build, or wrong configuration values, released	[1, p. 67]	1, 4
e	Firmware built for another controller	Scenario-360	2, 4
f	A defect added by the build tools	NEW	1
g	Code released without full testing	[1, p. 67]	1, 4

	Control	Source	Verified by
1	Firmware not judged good within the trial time is given up	Scenarios 430–450	TEST
2	The onboard computer refuses firmware that is damaged, older than the minimum version or built for another controller	Scenarios 360, 400, 420	TEST
3	The controller checks firmware again before it runs it	Scenario-610	TEST
4	The ground software checks every release before it is sent	Scenario-100	TEST
5	The operator can order a return to the previous firmware	NEW	DEMONSTRATION

Controls fail if: The judgement checks too little, or the trial ends before the defect shows; a check is made before the last transfer.

H-8 The controller stops working between contacts and is not brought back. CRITICAL. Software: CAUSE, CONTROL, MONITOR.
When: In any controller mode.
Effect: The controller's work stops until the ground acts at the next contact.

	Cause	Source	Controls
a	A bit error in running memory	Scenario-620; [1, p. 63]	1
b	The firmware hangs, deadlocks or overloads	[1, p. 63]	1, 4
c	A restart does not resume	[1, p. 63]	2
d	The restart takes longer than the time to the next failure	[1, p. 64]	**NONE**
e	The controller misbehaves when no command arrives for a long time	[1, p. 65]	2
f	The onboard computer misses the silence, or its watching program has crashed	Scenarios 520, 530	3, 5
g	The same defect is in every firmware copy, so every restart and the restore bring it back	NEW	**NONE**

	Control	Source	Verified by
1	The controller restarts on its own and resumes	Scenario-620	TEST
2	The onboard computer notices the silence, tries to bring the controller back and records it	Scenario-520	TEST
3	A crashed onboard computer program restarts and resumes	Scenario-530	TEST
4	New firmware that hangs is given up	Scenario-430	TEST
5	The controller keeps working while the whole onboard computer restarts	Scenario-535	TEST

Controls fail if: The controller's restart and the onboard computer's watch fail from one fault.

H-9 The ground is shown a false state of the System. CRITICAL. Software: CAUSE, CONTROL.
When: At every contact.
Effect: The operator decides on false information; a harmful command (H-6) or a missed fault (H-10) follows.

	Cause	Source	Controls
a	Wrong time-tags	[1, p. 64]	**NONE**
b	Old reports shown as current	[1, p. 63]	1
c	A report changed in storage or in transfer	[1, p. 62]	2
d	"Done" or "already done" answered for a command that did not run	NEW	**NONE**
e	The ground software stores or shows a report wrongly	[1, p. 62]	3
f	After a fall-back, the ground does not know which firmware runs	NEW	4

	Control	Source	Verified by
1	The onboard computer time-tags every report	Scenario-110	TEST
2	Every report is checked for damage on arrival	NEW	TEST
3	The ground software keeps every report received	[5, Sec. 3.2]	INSPECTION
4	After a fall-back, the next reports tell the ground which firmware runs	Scenarios 430–460	TEST

Controls fail if: The onboard clock is wrong; a report is damaged before its check.

H-10 A fault between contacts never reaches the ground. CRITICAL. Software: CAUSE, MONITOR.
When: Between contacts.
Effect: The ground does not know of the fault; hidden faults pile up, such as a damaged last-resort firmware (H-1).

	Cause	Source	Controls
a	The fault is not detected	[1, p. 66]	**NONE**
b	No event report is made	[1, p. 65]	**NONE**
c	The report is lost to full storage, a crash or a power loss	Scenarios 210, 530, 535	1, 2, 3
d	The ground loses it	NEW	4

	Control	Source	Verified by
1	Event reports are kept when storage is full	Scenario-210	TEST
2	No report is lost without the ground being told	O-5	TEST
3	Stored reports survive a power loss and a restart, including a restart of the whole onboard computer	Scenario-535	TEST
4	The ground software keeps every report received	[5, Sec. 3.2]	INSPECTION

Controls fail if: Fault detection depends on the part that has failed.

H-11 A valid command is not acted on, and the ground is not told. CRITICAL. Software: CAUSE, CONTROL.
When: While a command is checked, held or run.
Effect: The operator believes an action was taken that was not, for example a cancelled update (Scenario-470).

	Cause	Source	Controls
a	A full buffer drops the command	[1, p. 63]	1
b	A wrong onboard clock expires it	Scenario-340	2
c	A scheduled command is lost at a restart	Scenario-130	3
d	Data-link traffic blocks it	[1, p. 63]	1
e	Its answer is lost	Scenario-350	1

	Control	Source	Verified by
1	Every command gets two answers, received and then done with the result	Scenario-120	TEST
2	A refusal carries its reason	Scenarios 310–340	TEST
3	Scheduled commands survive a restart	NEW	TEST

Controls fail if: The answers are lost with the command.

H-12 A test function acts outside a test. CRITICAL. Software: CAUSE.
When: In operation.
Effect: Power is cut, a data-link connection is broken or the radio link is degraded in operation; each is a cause of H-1 or H-4.

	Cause	Source	Controls
a	Fault-injection functions stay in the operating software	[1, p. 64]	1
b	A wrong command, or one not from the operator, starts them	NEW	1

	Control	Source	Verified by
1	Test functions serve verification only and are not part of the operating software	Capability-5	INSPECTION

Controls fail if: A bench build is released for operation [1, p. 67].

7.0 Must-Work and Must-Not-Work Functions

The hazard analysis identifies the functions that must work to prevent a hazard, and those that must not work because they would cause one [1, App. A.1.2, p. 62]. Each holds only in the state named under When, and each must be covered by the requirements [7, App. F.2, p. 347].

Function	Type	When	Hazards
Return to the previous firmware when new firmware fails	MUST WORK	DURING A TRIAL	H-1, H-7
Run the last-resort firmware when the previous one fails	MUST WORK	AT A FALL-BACK	H-1
Restore through the reprogramming link	MUST WORK	NO FIRMWARE RUNS	H-1, H-2
Check every firmware image before it runs	MUST WORK	AT EVERY START	H-1, H-2, H-7
Refuse firmware and commands not from the operator	MUST WORK	ALWAYS	H-2, H-5
Accept the clock correction	MUST WORK	ALWAYS	H-4
Bring a silent controller back	MUST WORK	BETWEEN CONTACTS	H-8
Detect faults, and keep and send down every event report	MUST WORK	BETWEEN CONTACTS	H-8, H-10
Answer every command	MUST WORK	ALWAYS	H-11
Tell the ground which firmware runs	MUST WORK	AFTER A FALL-BACK	H-9
Replace a disclosed means of proving origin	MUST WORK	ON OPERATOR ORDER	H-2, H-5
Write over the previous or last-resort firmware	MUST NOT WORK	ALWAYS	H-1
Start failed firmware again without an operator order	MUST NOT WORK	AFTER A FALL-BACK	H-1
Run firmware not from the operator	MUST NOT WORK	ALWAYS	H-2
Run a command not from the operator	MUST NOT WORK	ALWAYS	H-5
Run a dangerous command without a valid arm	MUST NOT WORK	ALWAYS	H-6
Run a command twice, past its expiry, or at the wrong time	MUST NOT WORK	ALWAYS	H-6
Confirm firmware not judged good	MUST NOT WORK	DURING A TRIAL	H-7
Inject faults	MUST NOT WORK	OUTSIDE A TEST	H-12
Accept a new means of proving origin not proven with the current one	MUST NOT WORK	ALWAYS	H-4, H-5

Legend. MUST WORK: must happen, to prevent the hazards listed. MUST NOT WORK: must never happen, as it causes the hazards listed. When: ALWAYS; AT EVERY START,

whenever firmware starts; DURING A TRIAL; AT A FALL-BACK, when the controller returns to older firmware; AFTER A FALL-BACK; NO FIRMWARE RUNS; BETWEEN CONTACTS; ON OPERATOR ORDER; OUTSIDE A TEST.

A Top-Down Analysis

Figure 1 develops the worst hazard, H-1. Each box states what fails and when [4, p. 50].

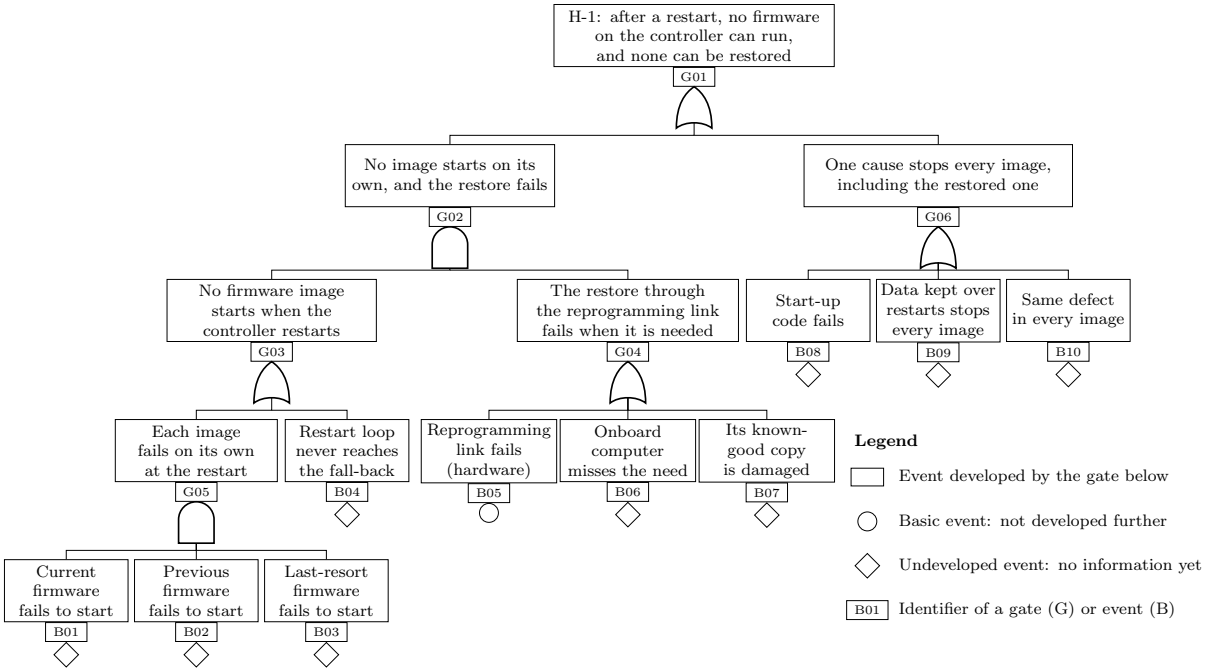


Figure 1: Fault tree of H-1. Symbols as in [4, Table 4-1, p. 34] and [4, Fig. 4-1, p. 33].

H-1 has three kinds of minimal cut set: one event (B08, B09 or B10); two events (B04 with one of B05 to B07); four events (B01, B02 and B03 with one of B05 to B07). By assumption 1 (Section 3.3), B08, B09 and B10 also stop the image the restore puts back, so each causes H-1 alone.

Section 4.0 gives the same count for every catastrophic and critical hazard.

B Bottom-Up Analysis

Before a design exists, the components examined are the System's functions [3, Sec. 4]. Failures are listed in an events table and a data table [3, Sec. 5.3]: each function for each way it can fail, and each item of data for each way it can be bad. Each failure is traced to the hazard it leads to.

Function	HALT	OMISSION	INCORRECT	TIMING	UNWANTED
Telemetry (Capability-1)					
Make health and event reports	H-10, H-15	H-10, H-15	H-9	H-9	H-4, H-15
Time-tag and store reports	H-10, H-15	H-10, H-15	H-9	H-9	H-15
Send reports down	H-16	H-10, H-15	H-9	H-16	H-16
Keep received reports on the ground	H-10, H-15	H-10, H-15	H-9	NONE	NONE
Command handling (Capability-2)					

Function	HALT	OMISSION	INCORRECT	TIMING	UNWANTED
Prepare and check commands on the ground	H-11	H-6	H-6	H-6	H-6
Check commands onboard	H-11	H-5, H-6	H-6, H-11, H-17	H-4, H-6, H-11	NONE
Hold armed and scheduled commands	H-11	H-11	H-6	H-6	H-6
Run commands	H-8, H-11	H-11	H-6	H-6	H-5, H-6
Replace the means of proving origin	H-4	H-2, H-5	H-4	H-4	H-4, H-5
Answer and log commands	H-9	H-11	H-9, H-17	H-16	H-9
Fault handling (Capability-3)					
Detect faults	H-8, H-10	H-8, H-10	H-10, H-13	H-8	H-13
Respond to faults	H-1, H-8	H-8	H-1, H-4, H-13	H-8	H-13
Update (Capability-4)					
Prepare and check firmware releases	NONE	H-7	H-3, H-7	NONE	H-7
Receive and check firmware	H-14	H-1, H-2, H-7	H-2, H-3, H-7, H-14	H-7	H-14
Install firmware and switch	H-1	H-14	H-1	H-1	H-6
Judge and confirm new firmware	H-13	H-13	H-3, H-7	H-7, H-13	H-7
Fall back to working firmware	H-1, H-8	H-1, H-8	H-1, H-7	H-8, H-13	H-13
Restore through the reprogramming link	H-1	H-1	H-1, H-2, H-7	H-1	H-13
Test (Capability-5)					
Inject faults	NONE	NONE	NONE	H-12	H-12

Legend: event failure type (types from [7, App. D.4.8, p. 336]). HALT: the function stops abnormally. OMISSION: the function does not happen. INCORRECT: it happens with wrong logic or a wrong result. TIMING: it happens at the wrong time or out of sequence. UNWANTED: it happens when it is not expected. NONE: leads to no hazard.

Data	OUT OF RANGE	MISSING	OUT OF SEQUENCE	OVERWRITTEN	WRONG
Command	H-6	H-11	H-6	H-6	H-6
Arm state	H-6	NONE	NONE	H-6	H-6
Firmware image	H-7	H-14	H-7, H-14	H-1, H-7	H-2, H-7
Report	H-9	H-10, H-15	H-9	H-10, H-15	H-9
Onboard time	H-4, H-6, H-9	H-6, H-9, H-11	H-6, H-9	H-6, H-9	H-4, H-6, H-9, H-11, H-13
Kept data	H-1, H-3	H-1, H-13	NONE	H-1, H-3, H-13	H-1, H-3, H-13
Means of proving origin	NONE	H-4	NONE	H-4	H-4

Legend: data failure type (types from [7, App. D.4.8, p. 335]). OUT OF RANGE: a value outside its possible range. MISSING: the data is absent. OUT OF SEQUENCE: the data arrives or is used in the wrong order. OVERWRITTEN: valid data is replaced. WRONG: the data is valid in form but incorrect. NONE: leads to no hazard. Kept data: the counts, flags, minimum version and controller identity kept over restarts.

The effect of each failure is the hazard in its cell; its severity, detection and compensation are those of that hazard. Every failure leads to a hazard of Section 4.0, or to none.

C Glossary

Common cause A cause that defeats several controls at once; it can negate a redundancy [4, Sec. 1.3, p. 3].

Control Hazard control: "means of reducing the risk of exposure to a hazard" [1, p. 13].

Fault tolerance "The ability to perform a function in the presence of any of a specified number of coincident, independent failure causes of specified types" [9, pp. 18–19].

Hazard "A state or a set of conditions, internal or external to a system that has the potential to cause harm" [1, p. 13].

Minimal cut set "The smallest combination of basic events that result in the top event" [4, Sec. 1.3, p. 3].

D Open Items

The values below are not yet known.

TBD-1 Project name and document reference.

The author accepted the severity rule, the meaning of each severity category and the full safety effort on 10 October 2026; and the fault tolerance rule, the two assumptions of this analysis (Section 3.3) and the hazard list with the severity of each hazard (Section 4.0) on 11 October 2026. No proposal awaits acceptance.